

Civil Engineering Materials (Mid Sem Exam- CLOSED BOOK)

Course No: CE F230

Date: 14-10-2023 (Saturday)

Duration: 9:00-10:30 AM, Part A + Part B = 90 mins.

Max. Marks: 50

Instructions:

1. Part B needs to be answered in a separate answer book.
2. Answer the questions in the same sequence as in the question paper.
3. Assume specific gravity of cement and fly ash as 3.15 and 2.1, respectively, wherever required.
4. Use Bogue's equations wherever required as given below:

$$A/F \geq 0.64$$

$$C_3S = 4.07(CaO) - 7.60(SiO_2) - 6.72(Al_2O_3) - 1.43(Fe_2O_3) - 2.85(SO_3)$$

$$C_2S = 2.87(SiO_2) - 0.75(C_3S)$$

$$C_3A = 2.65(Al_2O_3) - 1.69(Fe_2O_3)$$

$$C_4AF = 3.04(Fe_2O_3)$$

$$A/F < 0.64$$

$$C_3S = 4.07(CaO) - 7.60(SiO_2) - 4.48(Al_2O_3) - 2.86(Fe_2O_3) - 2.85(SO_3)$$

$$C_2S = 2.87(SiO_2) - 0.75(C_3S)$$

$$C_3A = 0$$

$$C_4AF = 2.1(Al_2O_3) + 1.70(Fe_2O_3)$$

1. Determine the initial porosity of binder (Portland Pozzolan cement) paste when the water to binder ratio is 0.40. The fly ash content is 30% of the total binder by weight. (2 marks)
2. The dimension of an aggregate particle is 6.4 mm x 11.2 mm x 19 mm. Classify the aggregate according to IS 2386: 1963 (2 marks)
3. Clearly explain differences between true SG, ASG, BSG_{OD}, BSG_{SSD} with formula and figure. (4 marks)
4. Explain with neat sketch and proper labeling, the hydration of a cement grain with time. (4 marks)

(P.T.O)

5. The oxides contents (%) of two cements are given below.

	Oxide	Cement A	Cement B
S	Silica	22.4	25
L	Lime	68.2	61
F	Iron	0.3	3
A	Alumina	4.6	4
SO ₃	SO ₃	2.4	2.5
	Free lime	3.3	1

- (a) Using Bogue's equation, determine the major chemical compounds of these cements and highlight the expected performance of the concrete mixes of a given composition when these cements were used, with reference to the following: (i) rate of strength development; (ii) sulfate attack [2+1+1]
- (b) Assuming the heat of hydration of C_3S , C_2S , C_3A , and C_4AF are 500, 260, 860, and 420 kJ/kg, respectively; determine the heat of hydration for these cements [2]
- (c) If these cements are used for a construction of a pile cap, assuming no loss of heat from the core of the pile cap, determine the temperature rise when these cements were used. Assume the specific heat of water is 4.2 kJ/kg/ °C and that for the solids is 0.22 times the specific heat of water. The mix proportion of the concrete is 1:2:4:0.50 (Cement: Fine aggregate: Coarse aggregate: Water) [1+1]
6. The following results are obtained using sieve analysis on fine and coarse aggregate. Plot the grading curves (in the graph provided) and determine the fineness modulus for these aggregates (fine, coarse and combined) if the fine and coarse aggregate are combined at 1:2 ratio. [10]

Coarse Aggregate		Fine Aggregate	
Sieve Size(mm)	Wt. Retained (g)	Sieve Size (mm)	Wt. Retained (g)
76	0	4.75	15
38	25	2.36	45
19.0	95	1.18	65
9.5	380	0.60	85
4.75	100	0.30	125
		0.15	50
		Pan	15

End of Part B